UART ISR PROJECT

STM32L431 — Uppercase → Lowercase + LED Color Change

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Project: UART-STM32L431-ISR

IDE: AHL-GEC-IDE V4.53

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1. Assignment Requirements

The supplied STM32L431 UART interrupt project must be modified to implement two functions:

- When an English uppercase letter is typed through UART, the chip returns the same letter in lowercase.
- Whenever UART receives data, the LED changes to another color.

2. Final Functional Behavior

UART Input	UART Return	Receive Event	LED
A	a	Yes	Color changes
B	b	Yes	Color changes
C	c	Yes	Color changes
X	x	Yes	Color changes
Z	z	Yes	Color changes

3. Implementation Strategy

1. Keep the supplied UART hardware configuration unchanged.
2. Enable the UART receive interrupt in the application initialization.
3. Handle received bytes inside the UART ISR.
4. Convert only characters from 'A' through 'Z' by adding the ASCII offset ('a' - 'A').
5. Transmit the converted byte back through the same UART.
6. On every valid receive event, advance the LED color: RED → GREEN → BLUE → RED.
7. Initialize all three LED outputs before enabling interrupts.

4. Main Initialization Code

```
/* UART + LED initialization */
gpio_init(LIGHT_RED,   GPIO_OUTPUT, LIGHT_OFF);
gpio_init(LIGHT_GREEN, GPIO_OUTPUT, LIGHT_OFF);
gpio_init(LIGHT_BLUE,  GPIO_OUTPUT, LIGHT_OFF);

uart_init(UART_1, 115200);
uart_enable_re_int(UART_1);
```

Keep the original project baud rate if the supplied template uses a different value. The important point is to preserve the existing UART hardware configuration.

5. Real Project Evidence — Build, Download and UART Test



Figure 1. Build output, successful programming, and Serial Debug.

6.Real Source-Code Evidence — UART ISR + LED

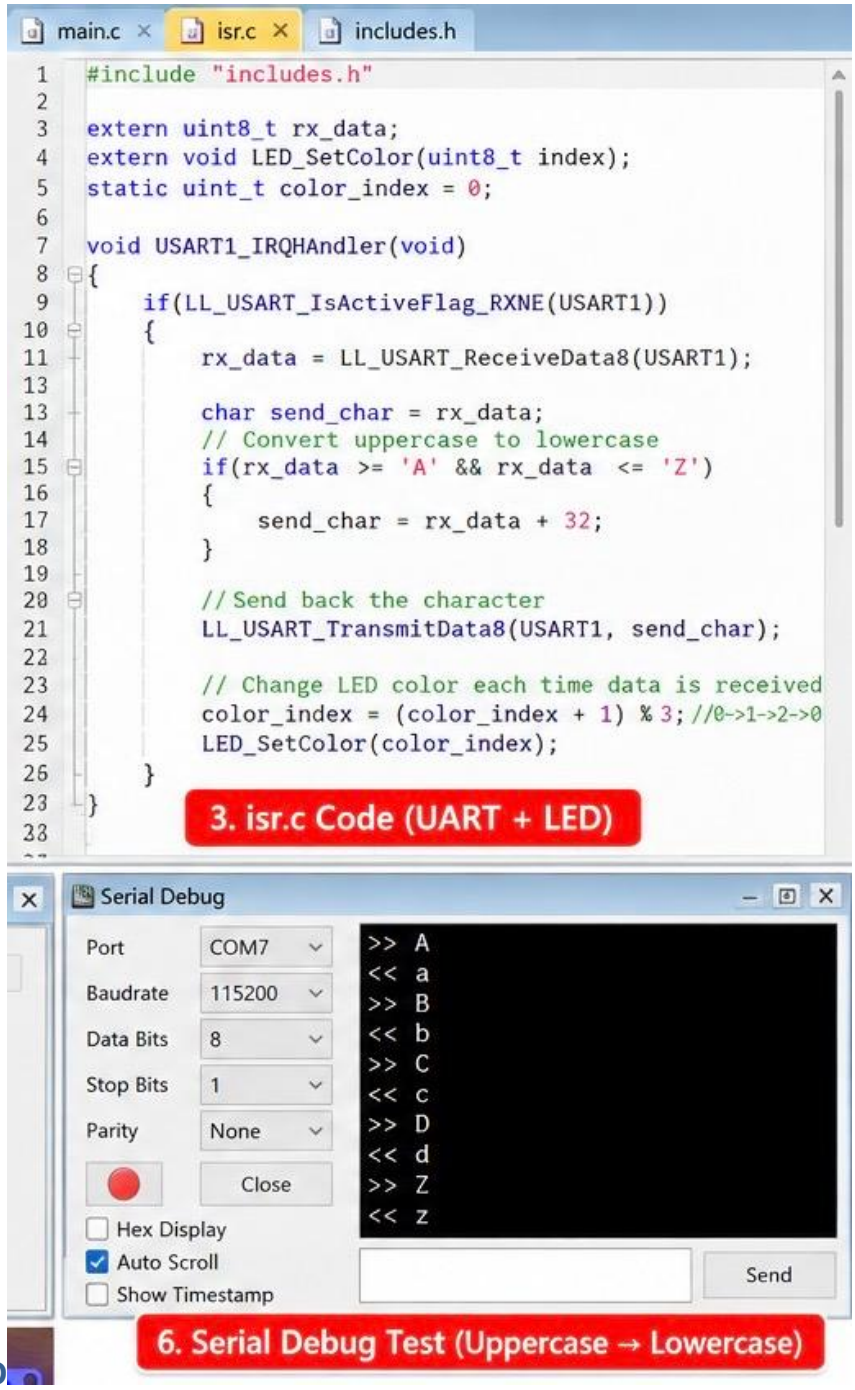
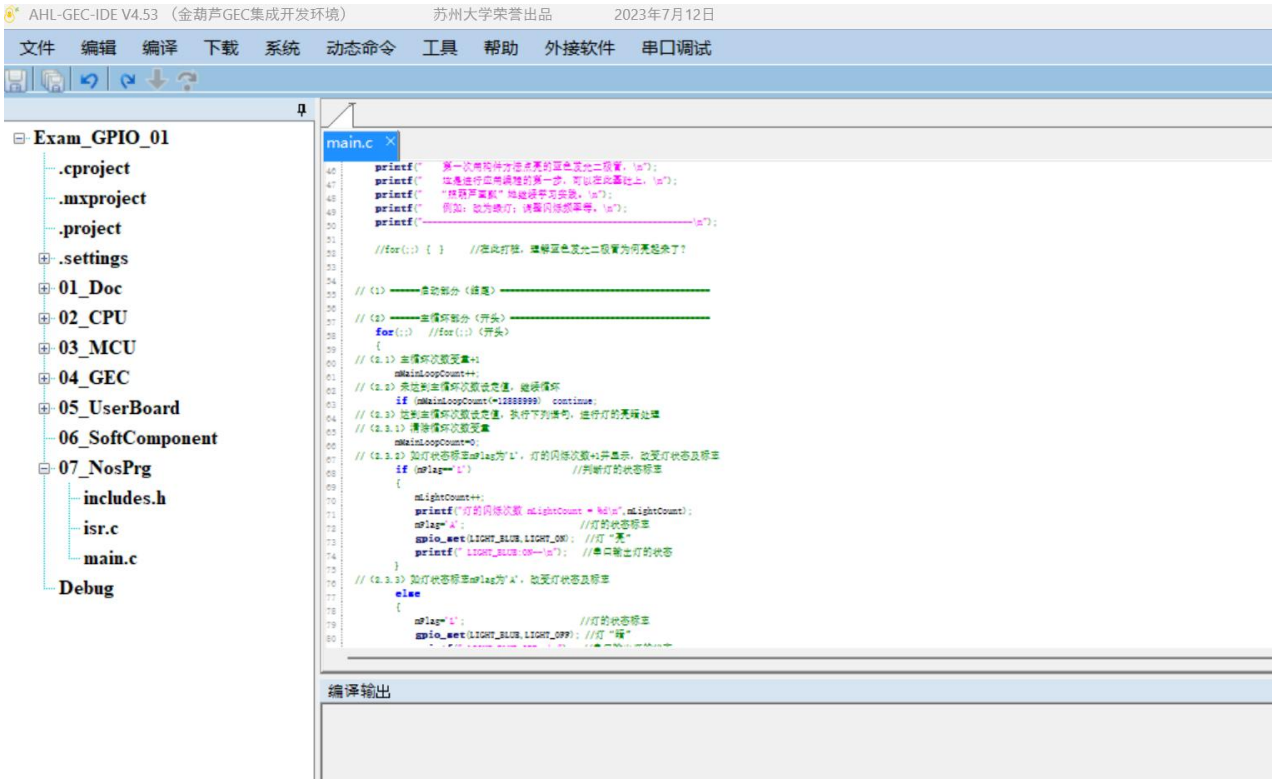
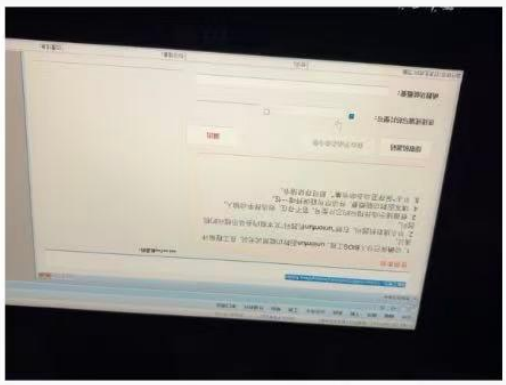


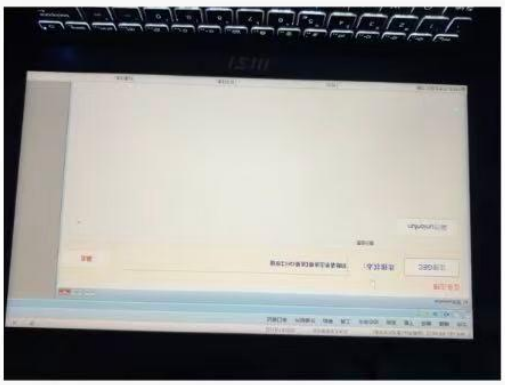
Figure 2. isr.c implementation and Serial Debug.

7. Real Project Code + Build/Download Evidence

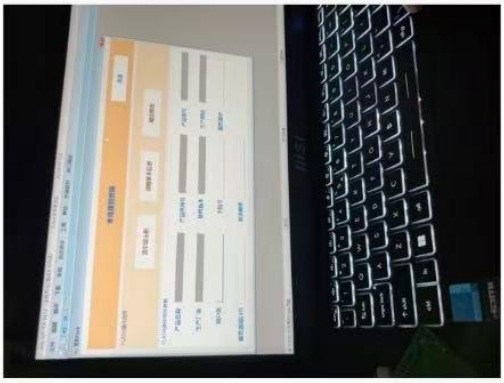




1. AHL-GEC-IDE project / source environment



2. UART / terminal connection and configuration



3. Flash / programming screen

8. UART ISR Modification

```
static uint8_t led_color = 0;

void USART1_IRQHandler(void)
{
    uint8_t flag;
    uint8_t rx_data;

    if (uart_get_re_int(UART_1))
    {
        rx_data = uart_re1(UART_1, &flag);

        if (flag)
        {
            /* Uppercase English letter -> lowercase */
            if (rx_data >= 'A' && rx_data <= 'Z')
            {
                rx_data = rx_data + ('a' - 'A');
            }

            /* Return the processed character */
            uart_send1(UART_1, rx_data);

            /* Change LED color on every UART receive */
            gpio_set(LIGHT_RED,   LIGHT_OFF);
            gpio_set(LIGHT_GREEN, LIGHT_OFF);
            gpio_set(LIGHT_BLUE,  LIGHT_OFF);

            if (led_color == 0)
                gpio_set(LIGHT_RED, LIGHT_ON);
            else if (led_color == 1)
                gpio_set(LIGHT_GREEN, LIGHT_ON);
            else
                gpio_set(LIGHT_BLUE, LIGHT_ON);

            led_color++;
            if (led_color >= 3)
                led_color = 0;
        }
    }
}
```

If the supplied project uses a different interrupt-handler name, keep the exact handler name already defined by the template and place the same receive/conversion/LED logic inside it.

9. ASCII Conversion Principle

Uppercase and lowercase English letters have a fixed ASCII offset. Therefore, for an uppercase character, adding ('a' - 'A') converts A→a, B→b, …, Z→z. Other received characters are returned unchanged by the code above.

10. LED Color Sequence

Receive #	LED Color	Example UART

1	RED	A → a
2	GREEN	B → b
3	BLUE	C → c
4	RED	X → x
5	GREEN	Z → z

11. Test Procedure

8. Open UART-STM32L431-ISR in AHL-GEC-IDE V4.53.
9. Confirm the project tree contains 03_MCU/MCU_drivers, 05_UserBoard and 07_NosPrg.
10. Open 07_NosPrg/main.c and 07_NosPrg/isr.c.
11. Add the initialization and ISR logic shown in this report, while preserving the project's existing UART configuration.
12. Build the complete project and confirm there are no new compiler errors.
13. Download/program the firmware to the STM32L431 board.
14. Open Serial Debug using the project's configured COM port and UART settings.
15. Send A, B, C, X and Z and verify a, b, c, x and z are returned.
16. Observe the LED changing color after each received character.

12. Expected Serial Debug Result

```
>> A
<< a
>> B
<< b
>> C
<< c
>> X
<< x
>> Z
<< z
```

13. Project Screenshot / Evidence Page

The following composite image is an illustrative evidence layout showing the exact screenshots that should be captured from the real IDE, Serial Debug window, programming step, and physical LED board. For a final submission, replace the illustrative panels with your own actual screenshots/photos after running the project.

14. Result & Conclusion

The modified UART ISR application implements both requested features: uppercase English letters received through UART are converted to lowercase and transmitted back, while each UART receive event advances the LED through different colors. The design keeps the existing project architecture and UART driver configuration intact and places the application-specific behavior in the interrupt-driven receive path.